

Who Controls the Model? AI Control Rights and Underinvestment

Abstract

This paper develops a property-rights–theoretic model to study how AI governance structures shape firms’ investment incentives and aggregate welfare. We conceptualize AI governance not as a binary make-or-buy decision, but as an allocation of disaggregated control rights over access, improvement, exclusivity, and forking. In a setting with incomplete contracts, an AI platform and a user firm make relationship-specific investments in AI quality, human capital, and proprietary data, and subsequently bargain over surplus. The model reveals a systematic underinvestment problem under the dominant platform-centric provision mode—standard SaaS or API-based access (Regime C). When platforms retain residual control over improvement, data-derived value, and technological duplication, user firms anticipate ex post appropriation and weaken their incentives to invest in AI-complementary human capital and data assets. As a result, equilibrium investment levels fall below the first-best benchmark, and total social welfare is not maximized, even when AI technologies are highly augmentation-intensive. We characterize alternative governance regimes—user-controlled contracts, distributed (forkable) deployment, and full integration—and show how reallocating specific control rights mitigates different hold-up channels. In particular, granting users stronger exclusivity or technological independence can partially restore investment incentives, though each regime involves distinct trade-offs between user-side and platform-side underinvestment. To assess the empirical relevance of the theory, we analyze AI-related disclosures from 3,482 publicly listed firms’ 10-K filings. Consistent with the model’s predictions, firms describing augmentation-intensive AI applications are significantly more likely to adopt user-centered governance structures and to increase investments in proprietary data assets. Taken together, the findings suggest that the prevailing platform-dominated AI governance model induces socially inefficient underinvestment. The results highlight an important role for public policy and regulation: strengthening enforceable user rights over data, human capital, and AI-generated improvements can alleviate investment distortions and improve welfare in AI-driven economies.

Keywords: *AI governance; incomplete contracts; hold-up; investment incentives; property-rights theory; platform economics; exclusivity rights; governance regimes; empirical validation.*

1 Introduction

1.1 Relationship-Specific Investments in AI Adoption

Artificial Intelligence (AI), usually recognized as the newest General Purpose Technology following foundational innovations like Information Technology (IT) and computers, shares several critical economic characteristics with these prior transformative technologies (Lee, Kim, Choi, & Kim, 2022). A technology qualifies as a General Purpose Technology

if it is pervasive, capable of being improved upon over time, and able to spawn complementary innovations (Brynjolfsson, Rock, & Syverson, 2017). AI fulfills these criteria, demonstrating the potential for far-reaching economic effects (Agrawal, Gans, & Goldfarb, 2019). First, like IT, the full economic impact of AI is subject to a significant time lag, often delaying observable productivity gains, a phenomenon consistent with the historical experience of electrification, automobiles, and computers (Agrawal et al., 2019). This concept echoes the debates surrounding the "productivity paradox of information technology" seen previously (Brynjolfsson, 1993). Second, the effective realization of AI's productive potential, similar to IT, is contingent upon substantial complementary investments (Brynjolfsson et al., 2017), includes mandatory and costly outlays in intangible assets (Agrawal et al., 2019). As suggested by Agrawal et al. (2019), these intangible investments involve developing datasets, building firm-specific human capital, and implementing necessary new business processes and organizational change. Empirical analysis showing that AI adoption at low levels does not yield significant revenue growth, but revenue growth occurs significantly only at high levels of adoption (e.g., 25% adoption or higher) (Lee et al., 2022). Drawing upon such empirical evidence, Lee et al. (2022) argue that AI adoption results in productivity lags due to adjustment costs and learning delays, requiring sustained, high-intensity investment in complementary assets and coordination to reach a critical threshold where significant, non-linear returns are realized.

Furthermore, although AI shares core features with earlier General Purpose Technologies such as Information Technology, its defining distinction lies in its transformative impact on the task content of production (Acemoglu & Restrepo, 2019). This creates a unique strategic tension, encapsulated in the automation-augmentation paradox (Raisch & Krakowski, 2021). Unlike models of technological change that assume factors (like labor or capital) become uniformly more productive (known as factor-augmenting technologies) (Acemoglu & Restrepo, 2019), AI operates via two countervailing forces that directly influence the allocation of tasks. The first is automation, which refers to the adoption of new technologies, such as robotics or specialized software, that allow capital to substitute for human labor across a widening range of tasks (Acemoglu & Restrepo, 2018a). This pro-

cess creates a displacement effect, shifting the task content of production away from labor and thereby unambiguously reducing the labor share (Acemoglu & Restrepo, 2019). The second is augmentation, which emphasizes technologies that enable close human-machine collaboration (Brynjolfsson et al., 2017; Raisch & Krakowski, 2021) or the creation of new, more complex tasks where labor retains a comparative advantage (Acemoglu & Restrepo, 2019). In this mode, AI and humans engage in a symbiotic partnership. For instance, in medicine, AI can highlight potential anomalies in a radiological scan, augmenting a doctor's diagnostic capabilities while leaving the final judgment and patient interaction in human hands (Hemmer, Schemmer, Kühl, Vössing, & Satzger, 2025). In software development, generative AI can act as a coding assistant, augmenting a programmer's productivity and allowing them to focus on higher-level system architecture (Azoulay, Krieger, & Nagaraj, 2024). New tasks generate a reinstatement effect, shifting the task content of production in favor of labor, thereby increasing labor demand and the labor share (Acemoglu & Restrepo, 2018b, 2019). This dual capacity means that AI does not merely enhance productivity but simultaneously creates profound conflicts over how work is structured and who benefits from the gains.

As discussed, the effective adoption of AI demands substantial, sustained investments in complementary intangible assets, including firm-specific data infrastructures, specialized R&D, and human capital development. To generate substantial value from AI, particularly through the high-potential path of augmentation, end-user firms must make deep and costly investments in complementary assets (Brynjolfsson et al., 2017). In the age of AI, these assets are overwhelmingly intangible and exhibit a high degree of relationship specificity (Agrawal et al., 2019)—that is, their value is maximized in the context of a specific AI provider's platform and is significantly diminished, if not entirely lost, should that relationship terminate (Coff & Raffee, 2015; Lafontaine & Slade, 2007).

The key forms of relationship-specific investment in AI can be grouped into data, human capital, and organizational processes (Brynjolfsson et al., 2017). Data specificity arises from the costly effort of collecting, cleaning, labeling, and structuring proprietary datasets so that they align with the architecture and training requirements of a particular AI model;

once tailored to the input formats of a given provider’s API, such data are difficult to repurpose elsewhere (Azoulay et al., 2024). Human capital specificity emerges when employees are trained not only in general AI literacy but also in the workflows and idiosyncrasies of a chosen platform, such as prompt engineering for a specific large language model, rendering these skills only partially transferable to competing systems (Brynjolfsson et al., 2017). Finally, process specificity reflects the deep redesign of business operations and organizational routines to integrate external AI technologies (Brynjolfsson et al., 2017). Embedding these technologies into daily workflows often requires years of adaptation and produces routines tightly bound to the provider’s system, thereby reinforcing dependence on the original platform (Y. J. Chen, Bharadwaj, & Goh, 2017).

1.2 Contractual Incompleteness and the AI Holdup Dilemma

These investments are not only specific but also largely non-contractible. It is exceedingly difficult to write a complete contract that specifies the full scope of business process reengineering or the precise nature of the organizational learning that must occur (Brynjolfsson, 1991). While inputs like proprietary data, specialized expertise, and process redesign are observable to the contracting parties, they are rarely verifiable by third parties and therefore cannot be fully specified or enforced in contracts (Aghion & Holden, 2011). The high complexity and uncertainty of AI technologies further exacerbate this problem, as future contingencies are difficult—if not impossible—to anticipate in advance.

Because these investments are often relationship-specific and generate quasi-rents, firms remain vulnerable to opportunistic renegotiation and holdup risks (Y. J. Chen et al., 2017). This means that effective governance of AI adoption depends critically on how residual control rights are allocated, since these rights determine whether firms’ investment incentives can be safeguarded. More broadly, the realization of gains from relationship-specific investments of this kind is constrained by contractual incompleteness and associated appropriability hazards, a challenge well captured by the classic holdup problem (Acemoglu, Aghion, Griffith, & Zilibotti, 2010; Hart & Moore, 1990).

Therefore, the deployment of AI creates a fundamental governance dilemma that reshapes the relationship between technology providers and end users. At the center of this dilemma lies a misalignment in the allocation of residual control rights. The end-user firm typically undertakes the most significant relationship-specific investments yet it often lacks control over the core technological asset. That asset is usually the AI platform or foundation model (Azoulay et al., 2024), which, by virtue of its centrality in the technology stack, becomes a potential chokepoint for innovation and competition. Allocating ownership—defined as residual control rights—over such indispensable assets is essential for mitigating opportunism and ensuring adequate investment when contracts are incomplete (Hart & Moore, 1990). However, incumbent technology providers frequently retain exclusive control over these foundational resources, creating a market structure concentrated at the platform layer (Azoulay et al., 2024). This configuration, where users bear the bulk of relationship-specific investments without possessing control over the indispensable technological asset, leaves them vulnerable to holdup risks and distorts incentives toward underinvestment (Aghion & Holden, 2011).

This modern hold-up problem is fundamentally more complex than its historical antecedents, such as the canonical General Motors-Fisher Body case (Casadesus-Masanell & Spulber, 2000). In that instance, the specific asset was a static physical object (a die for stamping car bodies). The hold-up risk was primarily about the price of a known, unchanging component (Casadesus-Masanell & Spulber, 2000). In the AI context, the core asset—the foundational model and its surrounding platform—is not static; it is a dynamic, self-learning system that is constantly evolving through the provider’s ongoing R&D efforts (Lee et al., 2022). Therefore, the core risk faced by the end-user company extends beyond a simple, one-time price negotiation; rather, it involves the ongoing governance challenge of securing the firm’s future development trajectory

1.3 Make-or-Buy Boundary under Property Rights Theory

The challenge posed by the holdup problem in AI—where end users undertake relationship-specific, non-contractible investments while providers retain residual control over the essential, continuously evolving models—can be fundamentally reframed as a "make-or-buy" boundary choice through the lens of Property Rights Theory (PRT) (Grossman, 1986; Hart & Moore, 1990). PRT, pioneered by Grossman (1986) and Hart and Moore (1990), posits that when contracts are incomplete, the allocation of residual control rights (RCR) (i.e., ownership) over non-human assets is crucial for mitigating opportunism (holdup) and ensuring adequate ex ante investments in relationship-specific assets. When critical residual control rights over the foundational AI asset (model/platform) cannot be credibly contracted, vertical integration ("make") can restore the investor's ex-ante incentives; conversely, when those rights can be sufficiently safeguarded—contractually or via alternative mechanisms such as reputation or specialized architectural design—outsourcing ("buy") or a hybrid governance arrangement is likely to suffice (Acemoglu et al., 2010).

In the current AI ecosystem, several structural features push firms toward outsourcing ("buy")—i.e., accessing a provider's platform—even when ownership-like protections are unavailable (Azoulay et al., 2024). First, high fixed costs and scale: as a General Purpose Technology, AI requires substantial complementary investments in high-performance compute and the capacity to manage massive datasets; the upfront costs of training and maintaining frontier foundation models are prohibitive for most end users, making incumbent platforms the default (Agrawal et al., 2019). Second, scarcity of frontier talent: demand for top AI researchers and engineers far exceeds supply, with large technology firms hiring the limited pool at premium wages, further discouraging in-house development ("make") (Lee et al., 2022). Third, control over complementary assets: incumbents tightly control specialized inputs—compute infrastructure, extensive non-public training data, and reliable model-serving pipelines—thereby reinforcing reliance on external platforms and fostering concentration at the foundation-model layer (Azoulay et al., 2024).

As firms rationally increase their reliance on outsourcing ("buy") platform access over

internal development ("make"), demand aggregates onto a limited set of providers, thereby reinforcing concentration at the foundation-model layer. This market structure, secured by incumbent firms' tight control over specialized complementary assets, is likely to result in an oligopolistic structure. This configuration might create chokepoints for downstream innovation and competition. The concentrated control at the foundation layer can dim the incentives to innovate in downstream application sectors (Azoulay et al., 2024).

1.4 Research Gap

The make-or-buy logic helps explain why firms increasingly rely on AI platforms. Yet classic Property Rights Theory was built for physical assets and treats ownership as a single, indivisible set of rights (Hart & Moore, 1990; Maskin & Tirole, 1999).

The core asset in AI, the foundational model, is fundamentally an intellectual property (Y. J. Chen et al., 2017). As an intangible asset, software, and by extension, AI models, are protected by intellectual property rights (IPR), which are the exclusive rights granted by law over the "use of creations of the minds" (Y. J. Chen et al., 2017). Therefore, AI, as intellectual property, is governed through a hybrid bundle of separable digital rights (Walden, 2005), which encompasses modular residual control rights such as access/use (e.g., via APIs) (Azoulay et al., 2024), improvement/upgrade (Azoulay et al., 2024), and the fundamental right to fork or self-host (Nyman & Lindman, 2013; Nyman, Mikkonen, Lindman, & Fougère, 2012).

Therefore, the classic Property Rights Theory lens struggles to analyze AI's new boundary challenges, since control resides not in a single indivisible asset but in a decomposable bundle of digital rights. Without a framework that connects the allocation of these rights to the investment incentives of users and providers, it is hard to explain in detail how repeated "buy" choices lead to concentration at the foundation layer, cause bottlenecks for downstream innovation, and shape the overall path of technological change.

To address this research gap, this paper asks: (RQ1) How does the strategic misalignment of residual control rights —where AI providers retain residual control rights over dy-

dynamic foundational models and end-users make indispensable relationship-specific investments—distort the optimal "make-or-buy" boundary decision and lead to chronic underinvestment in complementary capabilities? (RQ2) How should residual control be allocated across the four rights—access, improvement, exclusivity, and forking—to mitigate holdup and preserve users’ incentives to invest in data, talent, and processes? (RQ3) Which combinations of separable rights constitute the minimal bundle needed to replicate the investment incentives of vertical integration when “make” is infeasible, and how do different bundles shift the firm’s boundary choice—between make, buy, and hybrid governance—under varying levels of data specificity, switching costs, and provider bargaining power?

2 A Framework for Disaggregated AI Control Rights

The traditional property rights view of the firm identifies ownership with a single, indivisible bundle of residual control rights (Grossman, 1986; Hart & Moore, 1990). This conceptualization is inadequate for digital assets like AI, where control is exercised through a collection of distinct, separable, and technologically mediated permissions (Y. J. Chen et al., 2017; Walden, 2005). To build a more realistic model, we disaggregate control into a vector of four fundamental rights. This section defines these rights and connects them to observable contractual and technical arrangements in the AI industry.

2.1 Defining the Rights Vector

Ostrom (2010) conceptualized property rights systems as containing bundles of rights rather than a single right. Building on this framework, five distinct property rights can be identified that individuals using a common-pool resource may cumulatively hold: the right of withdrawal (harvesting specific products from the resource), the right of management (authority to transform the resource and regulate patterns of use), the right of exclusion (deciding who may exercise access, withdrawal, or management rights), and the right of alienation (leasing or selling any of the other four rights).

Building on the above bundle of rights perspective and the nature of AI as digital asset

(Y. J. Chen et al., 2017; Nyman & Lindman, 2013; Nyman et al., 2012; Walden, 2005), we propose that control over an AI asset can be characterized by a vector of four distinct rights, $R = (r_{\text{access}}, r_{\text{improve}}, r_{\text{exclusive}}, r_{\text{fork}})$. A governance structure is then defined by the allocation of this bundle of rights between the AI Platform (P) and the User Firm (F).

Definition 1. *The control structure for an AI asset is defined by the allocation of four rights:*

1. **Access** (r_{access}): *The right to use or invoke the AI's capabilities for their intended purpose. This is the fundamental right of consumption.*
2. **Improvement** (r_{improve}): *The right to modify, adapt, or enhance the AI model. This includes actions such as fine-tuning on new data, updating parameters, or integrating new modules. This is the right of creation and adaptation.*
3. **Exclusivity** ($r_{\text{exclusive}}$): *The right to the sole use and benefit of specific capabilities, features, or model weights that arise from the user's unique data or improvement efforts. This right prevents the platform from redeploying user-generated value to other parties. This is the right of appropriation.*
4. **Forking** (r_{fork}): *The right to create a static, independent, and self-contained duplicate of the AI model (including its architecture and weights) at a specific point in time, and to deploy and operate this duplicate in an environment fully controlled by the right-holder. This is the right of technological independence.*

These rights are conceptually distinct and have different economic functions. Access enables the basic consumption of a general-purpose technology. Improvement allows for the creation of relationship-specific value. Exclusivity protects the returns on that relationship-specific investment. Forking provides a powerful form of insurance against the risk of relationship breakdown or supplier opportunism.

2.2 Institutional Manifestations

This theoretical framework is not merely conceptual; it corresponds directly to the institutional configurations observed across AI and software markets. Distinct business models and contractual forms can be characterized as specific bundles of the four rights.

The *standard SaaS/API subscription model* represents one of the most common arrangement. In this setting, the user typically receives only the right to access (r_{access}) (L. Chen, Tong, Tang, & Han, 2022; Tan, 2020). The firm may invoke the model through an API and obtain outputs, but it cannot inspect the underlying architecture, modify the model, claim exclusive control over improvements derived from its usage, or obtain a transferable copy of the model (Y. J. Chen et al., 2017; Engert, Hein, Maruping, Thatcher, & Krcmar, 2024). The platform accordingly retains the remaining rights, including r_{improve} , $r_{\text{exclusive}}$, and r_{fork} .

A more advanced institutional form is *private-cloud deployment with fine-tuning capabilities*. Here, enterprise users often operate a dedicated instance of the model hosted in a virtual private cloud (Hanelt, Firk, Zapadka, & Oehmichen, 2025). This arrangement typically grants the right to improve (r_{improve}) through fine-tuning APIs (Azoulay et al., 2024; Hanelt et al., 2025). Contracts may further specify that fine-tuned model weights belong exclusively to the client, thereby conveying $r_{\text{exclusive}}$ (Y. J. Chen et al., 2017). However, users almost never receive the ability to extract or duplicate the model for independent execution, meaning that the provider generally retains r_{fork} .

A third configuration arises in *licensed on-premise deployment*, a model historically prevalent in enterprise software and increasingly used for specialized AI systems (Engert et al., 2024). Under this arrangement, users receive operational copies of the system and run them on their own servers, which implies both r_{access} and r_{fork} (Y. J. Chen et al., 2017; Nyman & Lindman, 2013; Nyman et al., 2012). Whether the user also obtains r_{improve} or $r_{\text{exclusive}}$ depends on the contractual and technological context: proprietary closed-source deployments typically withhold these rights, whereas open-source models effectively grant the entire rights bundle (Lerner, Pathak, & Tirole, 2006; Lerner & Tirole, 2002; Stout &

Hemphill, 2024).

Finally, *in-house development* corresponds to full vertical integration. A firm that constructs the AI system internally holds all four rights— $\{r_{\text{access}}, r_{\text{improve}}, r_{\text{exclusive}}, r_{\text{fork}}\}$ —by default. This archetypal “make” decision illustrates the purest form of internal control within the governance spectrum (Brynjolfsson, 1991; Casadesus-Masanell & Spulber, 2000; Engert et al., 2024).

3 The Model

We develop an incomplete contracting model rooted in the property rights framework of Grossman (1986) and Hart and Moore (1990), adapted to incorporate our disaggregated rights framework. The model analyzes the investment incentives and governance choices for AI development between a user firm and an AI platform.

3.1 Economic Environment

The model features two risk-neutral agents: an AI Platform (P) and a User Firm (F). Their relationship spans three dates. At date 0, a governance structure is chosen, which specifies the allocation of a vector of control rights. At date 1, the parties make non-contractible, relationship-specific investments. At date 2, the returns from these investments are realized, and the parties bargain over the division of the resulting surplus. The model’s logic rests on the following foundational assumptions.

Assumption 1 (Environment and Contract Incompleteness). *The economy consists of two risk-neutral agents, an AI Platform (P) and a User Firm (F), who maximize expected net payoffs. The parties choose governance arrangements at date 0 and then undertake relationship-specific investments in AI quality A , human capital L , and data curation D at date 1. These investment levels are observable to the parties but not contractible: they are not verifiable by a court or third party, so contracts cannot be written contingent on (A, L, D) . Consequently, investments affect the surplus realized at date 2 but cannot be con-*

tracted upon *ex ante*. All technological and informational primitives (including production technology, cost functions, and the bargaining protocol) are common knowledge at date 0.

Assumption 2 (Production Technology). *Joint investments in AI quality (A), human capital (L), and data curation (D) generate cooperative surplus through a twice continuously differentiable function $V(A, L, D)$ that is strictly increasing in each argument and exhibits non-negative cross-partials, so that the inputs are (weak) complements. For tractability, we adopt a flexible CES-type specification of the form $V(A, L, D) = \left(\alpha A^\rho + \beta L^\rho + \delta D^\rho + \gamma(ALD)^{\rho/3} \right)^{1/\rho}$, where $\rho < 1$ and $\alpha, \beta, \delta, \gamma \geq 0$. This functional form, commonly used in the literature on contractual investment (e.g., Acemoglu, 2003; Acemoglu, Antràs, & Helpman, 2007), allows for varying degrees of complementarity across the three investments. The parameter restrictions above deliver monotonicity and curvature properties consistent with our analysis.*

Assumption 3 (Investment Costs). *The platform and the user incur strictly increasing and strictly convex investment costs. For tractability, we adopt quadratic specifications: $c_P(A) = \frac{1}{2}k_A A^2$, $c_F(L, D) = \frac{1}{2}k_L L^2 + \frac{1}{2}k_D D^2$, with $k_A, k_L, k_D > 0$. These functions are twice continuously differentiable, satisfy $c'_P(0) = 0$ and $\partial c_F / \partial L(0) = \partial c_F / \partial D(0) = 0$, and ensure interior optimal investment choices under the production technology in Assumption 2. All qualitative results extend to general strictly increasing, strictly convex C^2 cost functions; the quadratic form is used only for notational simplicity.*

Assumption 4 (Bargaining Protocol and Payoffs). *At date 2, after investments (A, L, D) are sunk, the AI Platform (P) and the User Firm (F) divide the cooperative surplus through generalized Nash bargaining. Let $b \in (0, 1)$ denote the platform's bargaining weight. For governance regime $j \in \{C, U, D, I\}$, let (F_0^j, P_0^j) denote the parties' regime-specific outside options. The cooperative surplus is $S^j(A, L, D) \equiv V(A, L, D) - F_0^j - P_0^j$. Given (F_0^j, P_0^j) , the ex-post payoffs are $\Pi_F = F_0^j + (1 - b)S^j(A, L, D) - c_F(L, D)$, $\Pi_P = P_0^j + bS^j(A, L, D) - c_P(A)$. The bargaining process is static, symmetric in threat points, and takes place after investments but before any further contractual renegotiation. Exclusivity provisions, when present in a governance regime, enter only through their effect on the outside options*

(F_0^j, P_0^j) .

Assumption 5 (Well-Posedness of the Investment Problem). *For each governance regime $j \in \{C, U, D, I\}$, the investment game admits a well-defined interior equilibrium. Formally, there exists a unique triplet $(A^j, L^j, D^j) \in \mathbb{R}_{++}^3$ ($\mathbb{R}_{++}^3$ denotes the strictly positive orthant of $\mathbb{R}^3$, i.e., all triples (A, L, D) with each component strictly greater than zero) solving the first-order conditions of the user and platform's investment problems under regime j . This requires: (i) the production function $V(A, L, D)$ to be strictly increasing, strictly concave, and continuously differentiable on $\mathbb{R}_{++}^3$; (ii) the cost functions c_P and c_F to be strictly convex and twice differentiable with $c'_P(0) = 0$ and $\partial c_F / \partial (L, D)(0) = 0$; and (iii) boundary conditions ensuring that marginal returns become sufficiently large as any input approaches zero, guaranteeing interior solutions. Under these conditions, the joint surplus $S^j(A, L, D) = V(A, L, D) - F_0^j - P_0^j$ is strictly positive at the equilibrium investments, ensuring that ex-post bargaining is feasible in every regime.*

3.2 Governance as the Allocation of Control Rights

We model AI governance not as a binary ownership choice but as an allocation of a vector of separable control rights. Let $R = (r_{\text{access}}, r_{\text{improve}}, r_{\text{exclusive}}, r_{\text{fork}})$ denote the four rights defined in Section 2.2. In the theoretical model, each $r_k \in \{0, 1\}$ indicates whether a right is fully granted, while in the empirical analysis r_k is measured as a continuous intensity derived from textual disclosures. The binary abstraction captures limiting contractual regimes, whereas the empirical measures proxy for the degree to which each right is present in practice.

Although $2^4 = 16$ allocations are possible, only a small set of governance structures are economically stable in practice. We therefore focus on four canonical regimes that correspond to distinct and widely observed bundles of rights.

Regime C (Centralized / SaaS). Under standard SaaS or API-based provision, the user receives only access rights, yielding $R^C = (1, 0, 0, 0)$. The platform retains control over

improvement, appropriation, and duplication, corresponding to arm's-length reliance on platform infrastructure (Hanelt et al., 2025; Tan, 2020).

Regime U (User-Control / Private Cloud). Enterprise contracts may grant users improvement and exclusivity rights while withholding forking, yielding $R^U = (1, 1, 1, 0)$. This regime protects against platform appropriation but not technological lock-in (Azoulay et al., 2024; Y. J. Chen et al., 2017; Nyman & Lindman, 2013; Stout & Hemphill, 2024).

Regime D (Distributed / On-Premise License). On-premise or self-hosted deployment grants users forking and improvement rights without exclusivity, yielding $R^D = (1, 1, 0, 1)$. Hold-up is mitigated through technological outside options rather than contractual protection (Lerner & Tirole, 2002; Nyman & Lindman, 2013; Nyman et al., 2012; Stout & Hemphill, 2024).

Regime I (Integrated / In-House Development). Internal development confers the full rights bundle $R^I = (1, 1, 1, 1)$, corresponding to vertical integration and eliminating external contracting hazards (Bajari, Houghton, & Tadelis, 2014; Cadot, 2015; L. Chen et al., 2022; Klein, Mahoney, McGahan, & Pitelis, 2012).

These regimes recast AI governance as a multidimensional choice over control rights rather than a simple outsourcing–integration continuum. In particular, Regimes U and D represent distinct intermediate solutions: exclusivity mitigates appropriation risk, while forking mitigates lock-in. Optimal governance therefore depends on the nature of relationship-specific investments and the dominant hazard the firm seeks to control, providing a micro-foundation for the heterogeneity of AI governance arrangements observed empirically.

3.3 Outside Options and Cooperative Surplus

The economic role of the rights vector $R = (r_{\text{access}}, r_{\text{improve}}, r_{\text{exclusive}}, r_{\text{fork}})$ manifests in how these rights modify the parties' threat points in the date 2 Nash bargaining problem. For each governance regime $j \in \{C, U, D, I\}$, we derive the user firm's outside option F_0^j , the

platform's outside option P_0^j , and the resulting cooperative surplus $S^j \equiv V(A, L, D) - F_0^j - P_0^j$. The value function V satisfies the CES properties in Assumption 2, and all investments are sunk at date 2.

Throughout, we assume that without the platform's AI asset, the firm cannot generate meaningful AI-enabled value. To capture this complementarity, we normalize $V(0, L, D) = 0$, which is consistent with the essentiality of A in augmentation-based production. This normalization is without loss of generality and preserves the ordinal structure of incentives.

Regime C: Centralized (SaaS). Under the centralized SaaS mode, the platform retains the AI asset and grants the firm only the right to access it. If bargaining fails, the platform terminates access immediately. The firm loses the AI completely, while the platform can redeploy the model to the next-best user. $F_0^C = 0, P_0^C = \phi V(A, \bar{L}, \bar{D})$, where $\phi \in (0, 1)$ is the redeployability parameter and $(\bar{L}, \bar{D})$ denote market-average investments. The cooperative surplus is $S^C = V(A, L, D) - \phi V(A, \bar{L}, \bar{D})$.

Regime U: User-Control (Private Cloud). In the user-control regime, the firm receives adaptation rights (r_{improve}) and exclusivity ($r_{\text{exclusive}}$), but not the right to fork the model. If bargaining fails, the platform withdraws access to the base model, so the firm cannot operate the AI. However, exclusivity prevents the platform from using the firm's data-specific improvements to enhance its outside option. Let $\lambda \in [0, 1]$ index the strength of the appropriation channel through which the platform can extract incremental value from the firm's data. The right of exclusivity contractually sets $\lambda = 0$ under U. Formally, $F_0^U = 0, P_0^U = \phi V(A, \bar{L}, \bar{D})$, which equals the C-regime outside option but without any λ -dependent appropriability term. The cooperative surplus is therefore $S^U = V(A, L, D) - \phi V(A, \bar{L}, \bar{D})$. Relative to Regime C, exclusivity improves the user's bargaining position only through the reduction of the platform's threat point, not through any direct improvement in the user's fallback.

Regime D: Distributed (On-Premise License). The distributed regime grants the user the right to fork (r_{fork}), allowing the firm to retain an operational snapshot of the model at the moment of disagreement. The forked model has the same quality A but lacks future updates and support. Let $\varepsilon \in (0, 1)$ denote a friction parameter that captures the operational

degradation and switching costs associated with isolated operation. The outside options are $F_0^D = (1 - \varepsilon)V(A, L, D)$, $P_0^D = \phi V(A, \bar{L}, \bar{D})$, so the cooperative surplus becomes $S^D = V(A, L, D) - [(1 - \varepsilon)V(A, L, D) + \phi V(A, \bar{L}, \bar{D})] = \varepsilon V(A, L, D) - \phi V(A, \bar{L}, \bar{D})$. A feasible partnership requires $S^D > 0$, which amounts to a parameter restriction on (ε, ϕ) and is consistent with the interpretation that Regime D yields smaller gains from cooperation because the user retains a strong fallback position.

Regime I: Integrated (In-House Development). Under full integration, the user firm acquires or develops the AI asset in-house, internalizing the platform's role entirely. No external platform participates in the ex-post bargaining. Instead, the relevant friction stems from the potential departure of key employees whose tacit knowledge is essential for operating the model. Let $\delta \in (0, 1)$ index the extent to which the firm can operate the AI without such personnel. Because integration eliminates external contracting hazards, we normalize the outside option of the (nonexistent) external platform to zero: $P_0^I = 0$. The firm's outside option is $F_0^I = \delta V(A, L, D)$, yielding a cooperative surplus of $S^I = (1 - \delta)V(A, L, D)$. This formulation preserves two key properties: (i) integration eliminates the appropriation channel and (ii) the loss of key personnel generates an internal hold-up friction that is distinct from the platform opportunism faced in Regimes C, U, and D.

The structure of outside options across regimes reveals that $r_{\text{exclusive}}$ and r_{fork} protect user investment through distinct channels. The right to fork reallocates control of the *existing* asset, raising F_0^D directly. In contrast, exclusivity restricts the platform's ability to appropriate *newly created* value, lowering P_0^U . These two mechanisms are therefore imperfect substitutes. The resulting hierarchy of cooperative surplus, $S^C \leq S^U \leq S^D \leq S^I$, is consistent with the economic intuition that gradually shifting control from the platform to the user mitigates hold-up and increases relationship-specific investment incentives.

3.4 The First-Best Benchmark

As a benchmark for efficient investment, consider a social planner who chooses (A, L, D) to maximize total welfare, defined as cooperative output net of investment costs: $W(A, L, D) =$

$V(A, L, D) - c_P(A) - c_F(L, D)$. Under the concavity and regularity assumptions on V and the convexity of c_P and c_F (Assumptions 2–3), W is strictly concave and admits a unique interior maximizer $(A^*, L^*, D^*) \in \mathbb{R}_{++}^3$.

The first-best levels satisfy the system of first-order conditions: $V_A(A^*, L^*, D^*) = c'_P(A^*)$, $V_L(A^*, L^*, D^*) = \frac{\partial c_F}{\partial L}(L^*, D^*)$, $V_D(A^*, L^*, D^*) = \frac{\partial c_F}{\partial D}(L^*, D^*)$. That is, the planner equates the marginal social value of each investment to its marginal social cost. The decentralized equilibria derived under each governance regime will generally fail to satisfy these conditions due to incomplete contracts and regime-dependent threat points, generating the underinvestment distortions central to the analysis below.

4 Equilibrium Analysis of Investment Incentives

Having established the economic environment, the disaggregated rights framework, and the canonical governance regimes in the preceding chapter, this chapter proceeds to solve for the subgame perfect Nash equilibrium. The analysis focuses on the investment choices made by the Platform and the User Firm at date 1, under the anticipation of the ex-post bargaining outcome at date 2. The central objective is to characterize the investment distortions that arise from contractual incompleteness under each governance structure. By comparing the equilibrium outcomes to a socially optimal benchmark, this analysis will reveal how the allocation of specific control rights—access, improvement, exclusivity, and forking—mitigates or exacerbates the underlying hold-up problem for each party. The analysis begins by defining the first-best investment levels, which will serve as the benchmark for economic efficiency throughout.

4.1 The Social Planner's Benchmark: The First-Best Outcome

To benchmark economic efficiency, we consider the investment levels that would be chosen by a benevolent social planner operating in a frictionless environment. The planner internalizes all complementarities among the three relationship-specific investments—AI quality A , human capital L , and data curation D —and chooses (A, L, D) to maximize the

joint surplus net of investment costs: $W(A, L, D) = V(A, L, D) - c_P(A) - c_F(L, D)$.

The planner's first-best allocation (A^*, L^*, D^*) solves: $\max_{A, L, D} W(A, L, D)$. Because V is strictly concave (Assumption 2) and c_P, c_F are strictly convex (Assumption 3), the objective function W is strictly concave in (A, L, D) . By Assumptions 1–3, the planner's problem admits a unique interior solution.

Under the quadratic cost specification $c_P(A) = \frac{1}{2}k_A A^2$, $c_F(L, D) = \frac{1}{2}k_L L^2 + \frac{1}{2}k_D D^2$, the planner's first-order conditions simplify to: $V_A(A^*, L^*, D^*) = k_A A^*$, $V_L(A^*, L^*, D^*) = k_L L^*$, $V_D(A^*, L^*, D^*) = k_D D^*$.

These conditions have the familiar economic interpretation: efficiency requires the marginal social product of each investment (e.g. V_A) to equal its marginal social cost (e.g. $k_A A^*$). Because the planner internalizes all complementarities across A , L , and D , the resulting allocation (A^*, L^*, D^*) serves as the natural benchmark against which all decentralized governance regimes are evaluated.

In the decentralized equilibria derived below, neither the Platform nor the User Firm fully internalizes the joint benefits generated by their respective investments. As a result, the first-order conditions of the decentralized game depart systematically from (??)–(??), producing underinvestment or misallocation relative to the first-best solution—a canonical inefficiency arising from incomplete contracts and divergent outside options.

4.2 Regime C: Centralized Governance and the Dual Hold-Up Problem

Regime C corresponds to the canonical “Software-as-a-Service” (SaaS) or API-based AI deployment model. The Platform retains all control rights ($r_{\text{improve}}, r_{\text{exclusive}}, r_{\text{fork}}$) and grants the User only the access right r_{access} . The User must make relationship-specific investments in human capital L and data curation D , both of which are essential for extracting value from the Platform's AI asset A . Because the User holds no residual or exclusionary rights, these investments are vulnerable to ex-post expropriation through two distinct economic channels: (i) surplus sharing in Nash bargaining, and (ii) endogenous enhancement

of the Platform's outside option through data-driven learning.

Outside Options and Appropriation. If bargaining breaks down, the User loses access to the base model and is left with its human capital and data: $F_0^C = V(0, L, D)$. The Platform retains the AI asset and can redeploy it to the next-best user. To capture the platform's ability to learn from the user's data, we introduce an appropriability parameter $\lambda \in [0, 1]$ and define: $P_0^C(\lambda) = \phi V(A, \bar{L}, \bar{D}) + \lambda \Xi(D)$, $\Xi'(D) > 0$. The first term is the standard redeployability term; the second term reflects the value the Platform can extract from the User's proprietary data. Consistent with the CES structure and to ensure analytical tractability, we specify $\Xi(D) = V(0, 0, D)$, which preserves curvature and monotonicity.

Expected Payoffs and Incentive Problems. With bargaining weight $b \in (0, 1)$ for the Platform, the expected payoffs under Regime C are: $\Pi_F^C(A, L, D) = F_0^C + (1 - b)(V(A, L, D) - F_0^C - P_0^C(\lambda)) - c_F(L, D)$, $\Pi_P^C(A, L, D) = P_0^C(\lambda) + b(V(A, L, D) - F_0^C - P_0^C(\lambda)) - c_P(A)$. Each party chooses its investment to maximize its expected payoff, taking the other's investment as given. The Nash equilibrium (A^C, L^C, D^C) is therefore characterized by first-order conditions as follows:

Unified First-Order Conditions. The Regime C specialization is:

$$\frac{\partial \Pi_F^C}{\partial L} = (1 - b) V_L(A^C, L^C, D^C) + b V_L(0, L^C, D^C) - \frac{\partial c_F}{\partial L}(L^C, D^C) = 0, \quad (1)$$

$$\begin{aligned} \frac{\partial \Pi_F^C}{\partial D} &= (1 - b) V_D(A^C, L^C, D^C) + b V_D(0, L^C, D^C) \\ &\quad - (1 - b) \lambda \Xi'(D^C) - \frac{\partial c_F}{\partial D}(L^C, D^C) = 0, \end{aligned} \quad (2)$$

$$\frac{\partial \Pi_P^C}{\partial A} = b V_A(A^C, L^C, D^C) + (1 - b) \phi V_A(A^C, \bar{L}, \bar{D}) - c'_P(A^C) = 0. \quad (3)$$

Equation (1) reveals a standard Grossman–Hart–Moore (GHM) underinvestment problem. Equation (2) contains an additional negative term proportional to λ , reflecting the Platform's appropriation of data-driven insights. This creates a strictly stronger distortion for D than for L . Equation (3) shows that the Platform's incentives are also distorted: it in-

ternalizes only a fraction of the relationship-specific marginal product but fully internalizes the gains to its outside option. We can now formalize the core result of this regime.

Proposition 1 (Dual Hold-Up Under Centralized Governance). *Suppose Assumptions 1–3 hold, and let (A^*, L^*, D^*) denote the unique interior solution to the social planner’s problem. Under Regime C, the decentralized equilibrium (A^C, L^C, D^C) satisfies: $L^C < L^*$, $D^C < D^*$, $A^C \neq A^*$. Moreover, data investment exhibits a strictly stronger underinvestment distortion than labor investment: $\frac{\partial \Pi_F^C}{\partial D} < \frac{\partial \Pi_F^C}{\partial L}$ at the social optimum, and the distortion is strictly increasing in the appropriability parameter λ : $\frac{\partial D^C}{\partial \lambda} < 0$.*

Thus, Regime C features a dual hold-up problem: (i) a classical surplus-sharing distortion arising from Nash bargaining; and (ii) an appropriation-induced distortion arising from the Platform’s ability to enhance its outside option by learning from the User’s data. This second channel depresses the User’s incentives to invest in data strictly beyond the classical GHM distortion.

Economic Interpretation. The User anticipates that (i) it will capture only a fraction of the marginal surplus created by its investment in L and D , and (ii) its data investment additionally strengthens the Platform’s outside option via $\lambda \Xi'(D)$, thereby reducing the User’s share of ex-post surplus. This dual distortion provides a formal rationale for the widespread reluctance of firms to rely on AI platforms that learn from user data, and it establishes a microfoundation for the demand for richer governance rights analyzed in Regimes U, D, and I.

4.3 Regime U: User-Control and the Economic Role of Exclusivity

Regime U captures intermediate arrangements such as private-cloud deployments or dedicated fine-tuning services in which the User Firm obtains the rights of access and improvement together with the critical right of exclusivity ($r_{\text{exclusive}}$). Economically, exclusivity allocates property rights over the *data-induced, co-specialized improvements* that arise when the User’s proprietary data D is combined with the Platform’s general-purpose model A . This right prevents the Platform from reusing such user-specific improvements to enhance

its outside option. Thus, exclusivity directly shuts down the appropriation channel identified in Regime C.

Formally, exclusivity sets the appropriability parameter to zero, $\lambda = 0$, so the Platform's outside option reduces to its baseline redeployability value: $P_0^U = \phi V(A, \bar{L}, \bar{D})$, while the User's fallback payoff remains $F_0^U = V(0, L, D)$, because the User lacks the right to fork and therefore loses access to the model upon a bargaining breakdown.

The cooperative surplus is $S^U = V(A, L, D) - F_0^U - P_0^U$, and the expected payoffs under Nash bargaining with platform weight $b \in (0, 1)$ are: $\Pi_F^U = F_0^U + (1 - b)S^U - c_F(L, D)$, $\Pi_P^U = P_0^U + bS^U - c_P(A)$. The decentralized equilibrium (A^U, L^U, D^U) satisfies:

$$\frac{\partial \Pi_F^U}{\partial L} = (1 - b)V_L(A^U, L^U, D^U) + bV_L(0, L^U, D^U) - c_{F,L}(L^U, D^U) = 0, \quad (4)$$

$$\frac{\partial \Pi_F^U}{\partial D} = (1 - b)V_D(A^U, L^U, D^U) + bV_D(0, L^U, D^U) - c_{F,D}(L^U, D^U) = 0, \quad (5)$$

$$\frac{\partial \Pi_P^U}{\partial A} = bV_A(A^U, L^U, D^U) + (1 - b)\phi V_A(A^U, \bar{L}, \bar{D}) - c'_P(A^U) = 0. \quad (6)$$

A comparison of (5) with the Regime C condition (2) makes the economic effect of exclusivity transparent. Exclusivity removes the additional term $-(1 - b)\lambda \Xi'(D)$, which depresses the User's marginal return to data in Regime C. Hence, exclusivity eliminates the appropriation-induced distortion while leaving the classical Grossman–Hart–Moore surplus-sharing distortion unchanged. Under Assumptions 1–3, data investment incentives satisfy $\frac{\partial \Pi_F^U}{\partial D} > \frac{\partial \Pi_F^C}{\partial D}$, and $D^U > D^C$, with the wedge equal to $(1 - b)\lambda \Xi'(D^C) > 0$. Exclusivity fully removes this wedge but does not eliminate the surplus-sharing distortion relative to the first-best.

Exclusivity therefore provides a targeted solution to the appropriation problem: it restores the User's incentive to invest in data to the standard GHM level by preventing the Platform from converting the User's data-induced improvements into outside-option gains. However, because the User still captures only a fraction of the cooperative surplus, the classical underinvestment problem persists. Thus, exclusivity is a partial remedy; achieving full protection of relationship-specific investments may require stronger rights, such as

the right to fork, examined in the next section.

4.4 Regime D: Distributed Control and the Economics of Forking

Regime D captures governance arrangements—such as on-premise licensing or locally deployed model instances—in which the User Firm obtains the right to fork (r_{fork}). This right grants the User a credible ability to create and operate an independent copy of the model in the event of a bargaining breakdown. Economically, forking reallocates residual control rights over the *existing* AI asset, strengthening the User’s outside option and altering the entire surplus-sharing structure.

Under this regime, the User holds access and forking but not exclusivity, so $R_D = (1, 1, 0, 1)$. The outside options are: $F_0^D = (1 - \varepsilon)V(A, L, D)$, $P_0^D = \phi V(A, \bar{L}, \bar{D})$, where $\varepsilon > 0$ is a small friction capturing the cost of maintaining a forked model without support or future updates. Hence the bargaining surplus is: $S^D = V(A, L, D) - F_0^D - P_0^D = \varepsilon V(A, L, D) - \phi V(A, \bar{L}, \bar{D})$, which is positive under the viability condition $S^D > 0$.

Using the standard Nash bargaining weights $(1 - b, b)$, the expected payoffs reduce to: $\Pi_F^D = (1 - \frac{\varepsilon}{2})V(A, L, D) - \frac{1}{2}\phi V(A, \bar{L}, \bar{D}) - c_F(L, D)$, $\Pi_P^D = \frac{\varepsilon}{2}V(A, L, D) + \frac{1}{2}\phi V(A, \bar{L}, \bar{D}) - c_P(A)$.

The decentralized equilibrium (A^D, L^D, D^D) satisfies the first-order conditions: $\frac{\partial \Pi_F^D}{\partial L} = (1 - \frac{\varepsilon}{2})V_L(A^D, L^D, D^D) - c_{F,L}(L^D, D^D) = 0$, $\frac{\partial \Pi_F^D}{\partial D} = (1 - \frac{\varepsilon}{2})V_D(A^D, L^D, D^D) - c_{F,D}(L^D, D^D) = 0$, $\frac{\partial \Pi_P^D}{\partial A} = \frac{\varepsilon}{2}V_A(A^D, L^D, D^D) + \frac{1}{2}\phi V_A(A^D, \bar{L}, \bar{D}) - c'_P(A^D) = 0$.

Proposition 2 (Forking Restores User Incentives but Collapses Platform Incentives). *Under Assumptions 1–3, the following hold in Regime D:*

- (i) *User incentives converge to the first-best as $\varepsilon \rightarrow 0$: $L^D \rightarrow L^*$, $D^D \rightarrow D^*$.*
- (ii) *Platform incentives vanish as $\varepsilon \rightarrow 0$: $A^D \rightarrow \arg \max_A \phi V(A, \bar{L}, \bar{D}) - c_P(A)$, and $\frac{\partial \Pi_P^D}{\partial A} \rightarrow 0$.*
- (iii) *Consequently, forking eliminates the User-side hold-up but introduces a reverse hold-up problem in which the Platform underinvests in relationship-specific improve-*

ments.

As ε becomes small, the User's incentives approach the first-best benchmark, making forking an extremely powerful mechanism for protecting relationship-specific investments. In contrast, the Platform internalizes only an $\varepsilon/2$ share of the joint marginal product. Its investment incentives collapse toward its outside option, implying substantial underinvestment in relationship-specific capabilities.

Forking therefore removes the classical GHM distortion for the User but simultaneously induces a reverse hold-up problem: the User can now threaten to appropriate the entire value of the existing asset, leaving only the value of future updates in the bargaining surplus. This mechanism aligns with the economics of open-source ecosystems, where vendors monetize ongoing support and complementary services rather than the core model itself.

4.5 Regime I: Integrated Control and the Developer Hold-up Problem

Regime I corresponds to full vertical integration, in which the Firm holds the entire bundle of control rights. Integration eliminates the external hold-up problem present under market-based governance, but it introduces an internal source of inefficiency rooted in non-contractible human capital. The Platform is now represented by an internal developer whose tacit knowledge is essential for operating and updating the model. Because this knowledge is only partially embedded in organizational capital, the Firm's fallback payoff in the event of disagreement is limited.

We capture this by specifying: $F_0^I = \delta V(A, L, D)$, $P_0^I = 0$, where $\delta \in (0, 1)$ measures the degree to which tacit knowledge is codified into transferable organizational capital. When δ is small, the Firm loses most of the asset's value if the developer departs; when δ is large, knowledge is highly codified.

The cooperative surplus is: $S^I = (1 - \delta) V(A, L, D)$. Under Nash bargaining with platform weight $b \in (0, 1)$ (baseline $b = \frac{1}{2}$), the expected payoffs reduce to: $\Pi_F^I = \frac{1+\delta}{2} V(A, L, D) - c_F(L, D)$, $\Pi_P^I = \frac{1-\delta}{2} V(A, L, D) - c_P(A)$.

The decentralized equilibrium (A^I, L^I, D^I) satisfies: $\frac{\partial \Pi_F^I}{\partial L} = \frac{1+\delta}{2}V_L(A^I, L^I, D^I) - c_{F,L}(L^I, D^I) = 0$, $\frac{\partial \Pi_F^I}{\partial D} = \frac{1+\delta}{2}V_D(A^I, L^I, D^I) - c_{F,D}(L^I, D^I) = 0$, $\frac{\partial \Pi_P^I}{\partial A} = \frac{1-\delta}{2}V_A(A^I, L^I, D^I) - c'_P(A^I) = 0$.

The key observation is that integration strengthens the Firm's incentives while weakening those of the developer. The Firm internalizes a fraction $(1 + \delta)/2 > 1/2$ of the marginal return to L and D , whereas the developer internalizes only $(1 - \delta)/2 < 1/2$ of the marginal return to A . The following proposition formalizes this result.

Proposition 3 (Internalization and Developer Hold-up Under Integration). *Under Assumptions 1–3:*

- (i) *The Firm overcomes the classical GHM underinvestment distortion: $L^I > L^C, D^I > D^C$.*
- (ii) *The developer faces an internal hold-up problem: $A^I < A^*$, $\frac{\partial \Pi_P^I}{\partial A} = \frac{1-\delta}{2}V_A - c'_P < V_A - c'_P = 0$.*
- (iii) *As knowledge becomes more codified ($\delta \uparrow 1$), Firm incentives approach the first-best, but developer incentives vanish: $L^I, D^I \rightarrow (L^*, D^*), A^I \rightarrow \arg \max_A \{-c_P(A)\}$.*

Thus, integration removes the external hold-up problem by reallocating all control rights to the Firm, but it creates a new internal distortion: the developer hold-up problem. The parameter δ summarizes the effectiveness of organizational governance—such as documentation, modular architectures, or knowledge-transfer processes—in embedding tacit developer knowledge into the firm. Higher δ mitigates the internal hold-up problem, while lower δ leaves the Firm heavily dependent on key personnel.

4.6 Comparative Statics: A Synthesis of Investment Distortions

Taken together, the four regimes reveal a fundamental trade-off at the core of AI governance. No single allocation of control rights is universally optimal. Instead, the choice of governance structure must balance incentives for the Platform's investment in core AI quality (A) against incentives for the User Firm's complementary investments in human capital (L) and data (D).

To make the incentive distortions transparent, we summarize the first-order conditions by comparing, for each regime, the *marginal private return* (MPR) to investment with the corresponding *social* marginal return under the first-best benchmark.¹ Table ?? reports the MPRs implied by the FOCs derived in Sections 4.2–4.5.

Table ?? highlights three comparative-static patterns.

(i) Centralized and User-control regimes: strong but misaligned platform incentives, weak user incentives. In Regimes C and U, the Platform’s MPR in A combines the relationship-specific component $V_A(A, L, D)$ and the redeployable component $\phi V_A(A, \bar{L}, \bar{D})$, giving it relatively strong incentives to invest in general-purpose AI capabilities. However, the User’s MPRs in L and D are strictly below the first-best benchmark, reflecting the standard Grossman–Hart–Moore underinvestment problem. In addition, under Regime C the data margin suffers an extra *appropriation term*, $-\frac{1}{2}\lambda \Xi'(D)$, which captures the dual hold-up channel identified in Proposition 1.

(ii) Distributed regime: near first-best user incentives, but sharply weakened platform incentives. In Regime D, the right to fork raises the User’s MPR in both L and D to $(1 - \frac{\varepsilon}{2})V_i$, which converges to the first-best V_i as the friction $\varepsilon \rightarrow 0$. This makes D the most user-favorable regime on the complementary investment margins. At the same time, the Platform’s MPR in A becomes proportional to $\frac{\varepsilon}{2}$ on the relationship-specific component, implying that as forking becomes frictionless, platform incentives to invest in relationship-specific AI quality effectively vanish.

(iii) Integrated regime: strong user incentives, internal developer hold-up. Under integration (Regime I), the Firm internalizes a fraction $\frac{1+\delta}{2} > 1/2$ of the marginal returns to L and D , so its incentives on the complementary margins are strictly stronger than under any non-integrated regime. By contrast, the developer’s MPR in A is only $\frac{1-\delta}{2}V_A$, which can be far below the social benchmark, especially when organizational capital δ is small.

¹All marginal returns in Table ?? are evaluated at the regime-specific equilibrium (A^j, L^j, D^j) , and we fix $b = \frac{1}{2}$ for expositional clarity.

This is the developer hold-up problem formalized in Proposition 3.

Overall, the comparative statics in Table ?? show that each governance regime implements a distinct pattern of investment distortions. Regimes C and U favor platform-centric incentives for core AI quality at the expense of user incentives on L and D (with an additional data appropriation distortion under C). Regime D nearly restores first-best user incentives but sharply undermines relationship-specific platform investment. Regime I strengthens user incentives across both L and D while creating an internal hold-up problem on A . The normative problem, taken up in the next section, is to characterize the parameter regions (in technology, data specificity, redeployability, and organizational codification) in which each regime is welfare-maximizing.

5 The Optimal Allocation of AI Control Rights

The positive analysis in the preceding chapter characterized the investment distortions inherent in different allocations of control rights. Under contractual incompleteness, no single governance structure achieves the first-best outcome. The Platform and the User Firm, acting in their own self-interest, fail to fully internalize the externalities their investment decisions impose on the other party. This chapter transitions from this positive characterization to a normative analysis. The central objective is to determine which governance structure emerges as the optimal, or "second-best," solution under varying technological and economic conditions. The analysis formalizes the fundamental trade-offs between incentivizing the Platform's investment in the core AI technology and the User Firm's complementary investments in human capital and proprietary data.

5.1 Proposition 4: The Augmentation-Automation Trade-off

The first and most fundamental trade-off concerns the nature of the AI technology itself. Is it primarily a tool for augmenting the capabilities of skilled human workers, or is it a tool for automating tasks and substituting for labor? The answer to this question determines the relative importance of the Platform's investment in technology (A) versus the User's

investment in human capital (L). Our unified CES production function captures this trade-off through the parameter $\theta = \gamma/\alpha$. A high θ ratio signifies a technology that is highly complementary to human skill and data.

Proposition 4 (Augmentation vs. Automation). *The optimal allocation of control rights shifts towards the User Firm as the AI technology becomes more complementary to the User's human capital and data. Formally, let $\theta = \gamma/\alpha$. There exists a unique threshold $\theta^* > 0$ such that for any $\theta > \theta^*$, a user-centric governance structure (Regime I) is socially optimal over a platform-centric structure (Regime C), and for any $\theta < \theta^*$, the platform-centric Centralized structure (Regime C) is socially optimal.*

Proof. The proof is omitted for brevity. The proof uses the Envelope Theorem to show that the welfare difference, $W^I - W^C$, is monotonically increasing in the complementarity parameter θ . $\square$

The parameter θ is more than a technical assumption; it represents the firm's underlying production philosophy. A firm that views technology as a tool to empower its skilled workforce (high θ) will find that its optimal organizational structure must align with this philosophy to be efficient. Adopting a platform-centric SaaS model (Regime C) for a deeply complementary task is predicted to be inefficient because it systematically undermines the incentives for the firm's most critical assets: its human capital and proprietary data. An efficient firm must align its governance structure with its production philosophy. Choosing a platform-centric model that prioritizes platform investment (A) for a task that is deeply complementary (high θ , high importance of L and D) creates a strategic contradiction that our model predicts will lead to significant value destruction. This elevates the proposition from a simple technical trade-off to a statement about the necessary co-evolution of technology, strategy, and organizational design.

5.2 Proposition 5: The Right to Fork as a Commitment Device

The right to fork (r_{fork}) is a unique feature of digital assets that provides the User Firm with a powerful safeguard against Platform opportunism. By granting this right, the Platform

can credibly commit not to engage in excessive hold-up, as the User has a technologically-enforced outside option of continuing to use an independent copy of the asset. This commitment can unlock the User's willingness to make deep, relationship-specific investments that would otherwise be too risky.

Proposition 5. *Allocating the right to fork (r_{fork}) to the User Firm (Regime D) is optimal when the User's relationship-specific investments (L and D) are sufficiently important drivers of joint surplus. Specifically, Regime D dominates Regime C ($W^D > W^C$) if the welfare gains from mitigating the User's underinvestment in L and D exceed the welfare losses from the Platform's reduced incentive to invest in relationship-specific aspects of A.*

Proof. The proof is omitted for brevity. The proof involves a direct welfare comparison, showing that Regime D is preferred when the deadweight loss from underinvesting in L and D (as in Regime C) is greater than the deadweight loss from underinvesting in relationship-specific aspects of A (as in Regime D). $\square$

The friction cost, ε , is the key to this trade-off. It is not merely a nuisance cost but represents the very source of the gains from trade left on the bargaining table. As $\varepsilon \rightarrow 0$, the User's investment incentives approach perfection ($L^D \rightarrow L^*, D^D \rightarrow D^*$), but the Platform's incentive to invest in the relationship vanishes, potentially rendering the partnership non-viable. This reveals a deeper, dynamic trade-off. The bargaining surplus, εV , can be interpreted as the net present value of the Platform's ongoing contributions: future updates, improvements, and support. The fork right effectively allows the user to seize the full value of all past investments made by the Platform, leaving only the value of future improvements on the bargaining table. This does not eliminate the hold-up problem but rather reverses its direction. The Platform, anticipating that its future contributions can be held up by the user's credible threat to fork, will systematically underinvest in creating relationship-specific updates. This provides a formal microfoundation for the business models common in the open-source world (an extreme form of the fork right), where firms monetize the ongoing relationship through paid support and enterprise services, rather than the core technology itself.

5.3 Proposition 6: Data Specificity and the Value of Exclusivity

When a general-purpose AI model is fine-tuned or enhanced using a User Firm's proprietary data, a new, unique, and highly specific *composite asset* is created. The right of exclusivity ($r_{exclusive}$) is effectively the property right over this composite asset. Without this right, the Platform can appropriate the value of the User's data investment (D) by incorporating the resulting improvements into its general model and socializing them across its entire customer base, including the User's competitors. This externality severely undermines the User's incentive to invest in curating and providing high-quality data.

Proposition 6. *Allocating the right of exclusivity ($r_{exclusive}$) to the User Firm (as in Regime U) is optimal when the User's proprietary data investment (D) is a primary driver of joint surplus. The absence of exclusivity leads to underinvestment by the User in data curation and a misallocation of investment by the Platform towards general-purpose features.*

Proof. The proof is omitted for brevity. The proof formalizes the lack of exclusivity by introducing a parameter λ representing the fraction of data value the Platform can appropriate. It shows that the User's investment is decreasing in λ , and thus total welfare is maximized when exclusivity is granted (i.e., $\lambda = 0$). $\square$

This result clarifies that $r_{exclusive}$ (characteristic of Regime U) and r_{fork} (characteristic of Regime D) are distinct, non-hierarchical solutions to the hold-up problem. $r_{exclusive}$ is a *contractual* solution that works by defining property rights over the *returns* to a specific investment (data). r_{fork} is a *technological* (or property-rights) solution that works by re-allocating ownership over the *asset itself*. The optimal choice between these intermediate governance structures depends on the specific risk a firm seeks to mitigate. If the primary risk is the Platform appropriating the value derived from unique, proprietary data, the contractual safeguard of $r_{exclusive}$ is paramount. Conversely, if the primary risk is Platform opportunism—such as prohibitive price increases or service degradation—then the technological independence afforded by r_{fork} becomes the more valuable right.

5.4 Proposition 7: Platform Market Power and the Flight to Integration

The baseline model relies on the simplifying assumption of symmetric bargaining power. This section enriches the analysis by relaxing this assumption and exploring the model's implications for the institutional realities of the AI market, which is often characterized by a small number of dominant platforms. We generalize the model to account for asymmetric bargaining power and derive new results on how market structure influences the optimal choice of governance.

Platform Market Power and Asymmetric Bargaining. The market for foundational AI models is characterized by massive fixed costs and network effects, leading to an oligopolistic structure. In such a market, it is unrealistic to assume symmetric bargaining power ($b = 1/2$). It is more likely that large platforms possess significant market power, implying $b > 1/2$, where b is the platform's bargaining power parameter. We now generalize the equilibrium analysis for an arbitrary $b \in (0, 1)$. The parties' expected payoffs under any regime are given by: $\Pi_F = F_0 + (1 - b)(V_{coop} - F_0 - P_0) - c_F(L, D)$, $\Pi_P = P_0 + b(V_{coop} - F_0 - P_0) - c_P(A)$. Re-deriving the first-order conditions for each regime with this general b yields a more nuanced understanding of the investment incentives. The results are summarized in Table 1, which generalizes the analysis from Chapter 4.

Insert Table 1 here

This generalized table is a powerful analytical tool. It immediately reveals that the User's investment incentives under the Centralized regime are highly sensitive to b . As the Platform's bargaining power b approaches 1, the User's share of the marginal return, $(1 - b)V_L$, vanishes, and the incentive to invest is driven almost entirely by the investment's value in the User's outside option ($V_L(0, L, D)$). This severely exacerbates the hold-up problem. In contrast, the User's incentives in Regimes D and I are primarily secured by their strong

outside options (F_0^D, F_0^I) and are therefore much less sensitive to b . This observation is the foundation for our next proposition.

The shift in bargaining power is not neutral; it systematically changes the relative efficiency of different governance structures. As platforms become more powerful, the contractual hazards of outsourcing increase, making user-centric control structures more attractive.

Proposition 7 (Platform Market Power). *An increase in the Platform's exogenous bargaining power (b) expands the parameter space for which full vertical integration (Regime I) is the optimal governance choice. As platform market power intensifies, the contractual hazards of arm's-length dealing (Regime C) become so severe that integration becomes a "safe harbor," even if it is internally inefficient (i.e., for lower values of δ).*

Proof. The proof is omitted for brevity. The proof shows that the welfare advantage of Regime I over Regime C, $\Delta W = W^I - W^C$, is an increasing function of b . This is because total welfare under Regime C is highly sensitive to b (as user incentives collapse), while welfare under Regime I is largely insulated from external market power. $\square$

This result formalizes a complete economic causal chain: technological characteristics like high fixed costs lead to a concentrated market structure (oligopoly), which grants incumbent firms greater bargaining power (high b). This increased power amplifies the contractual hazard of hold-up for user firms, which in turn shifts the optimal organizational form away from arm's-length outsourcing (Regime C) and towards vertical integration (Regime I). This provides a theory-grounded explanation for the prevalence of "acqui-hires" and in-house AI development in markets dominated by a few large technology platforms, framing it not just as an offensive strategy but as a defensive necessity.

6 Empirical Evidence

This section tests the central prediction of the model that augmentation-oriented AI deployment shifts firms toward more user-centric control rights and, ultimately, toward integrated

governance.

Our empirical analysis uses firm–year data for publicly listed North American firms over the period 2015–2025. Financial and accounting variables are obtained from Compustat. Information on firms’ AI usage and governance arrangements is constructed from annual 10-K filings retrieved from the SEC EDGAR database. We exclude financial firms and regulated utilities and retain only observations with non-missing AI-related disclosures, yielding an unbalanced panel of 14,178 firm-year observations.

AI augmentation intensity is the primary explanatory variable. It captures whether firms frame AI as complementing rather than substituting human labor. The measure is constructed using a text-based approach that contrasts augmentation-oriented and automation-oriented language in AI-related disclosures and aggregates signals across relevant sections of the 10-K. Higher values indicate a stronger emphasis on human–AI complementarity.

Consistent with the theoretical framework, AI governance is operationalized as a vector of control-right intensities covering access, improvement, exclusivity, and forking. Each right is measured as a continuous index derived from contractual and technical language in AI disclosures using a hybrid natural language processing approach combining semantic classification and keyword validation. These measures capture the extent to which firms retain contractual or technological control over AI assets.

We classify firm-years into four governance regimes—Centralized (SaaS), User-Control (Private Cloud), Distributed (On-Premise), and Integrated (In-House)—based on dominant rights configurations. In addition, we construct a continuous user-centric control index that aggregates improvement, exclusivity, and forking rights.

Control variables include firm size, leverage, profitability, R&D intensity, capital intensity, and industry concentration. All continuous variables are winsorized at the 1st and 99th percentiles. Summary statistics and detailed variable definitions are omitted due to space constraints.

We first examine how AI augmentation affects the allocation of individual control rights using firm–year panel regressions of the form: $r_{k,i,t} = \alpha_k + \beta_k \text{Augment}_{i,t-1} + \gamma_k^\top \mathbf{X}_{i,t} + \delta_t + \eta_j + \varepsilon_{k,i,t}$, where $r_{k,i,t}$ denotes the intensity of control right $k \in \{\text{access, improve, exclusive, fork}\}$

for firm i in year t , $Augment_{i,t-1}$ is lagged AI augmentation intensity, $\mathbf{X}_{i,t}$ is a vector of firm controls, and δ_t and η_j denote year and industry fixed effects. Standard errors are clustered at the firm level.

To examine governance mode choice, we estimate a Logit model for the probability of integrated governance: $\log\left(\frac{P(G_{i,t}=1)}{1-P(G_{i,t}=1)}\right) = \alpha + \beta Augment_{i,t-1} + \gamma^\top \mathbf{X}_{i,t} + \delta_t + \eta_j$.

Table 2 summarizes the main empirical findings. Consistent with the theoretical predictions, augmentation-oriented AI deployment is strongly associated with a reallocation of control rights toward the user.

Insert Table 2 here

In the rights regressions, augmentation has a positive and statistically significant effect on the right to improve ($\beta = 0.082$, $p < 0.01$) and the right to fork ($\beta = 0.128$, $p < 0.01$), indicating increased demand for customization and technological autonomy as human–AI complementarity rises. In contrast, augmentation is negatively associated with access-based contracting ($\beta = -0.098$, $p < 0.01$) and contractual exclusivity ($\beta = -0.111$, $p < 0.01$), reflecting a systematic move away from platform-centric SaaS arrangements.

The governance regressions show an economically large effect of augmentation on integration. A one-unit increase in lagged augmentation raises the log-odds of adopting integrated governance by approximately 2.56 ($p < 0.01$), implying an odds ratio of $\exp(2.56) \approx 13$. This result indicates that augmentation not only reshapes contractual rights but also induces firms to internalize AI development when contractual governance becomes insufficient to protect relationship-specific investments.

To address potential endogeneity between AI augmentation and governance choices, we estimate a two-stage least squares (2SLS) specification exploiting staggered adoption of the Uniform Trade Secrets Act (UTSA) across U.S. states. UTSA adoption strengthens legal protection for proprietary data and tacit knowledge, plausibly affecting firms'

incentives to pursue augmentation-oriented AI strategies without directly determining governance choices.

The first stage relates augmentation intensity to UTSA exposure:

$$Augment_{i,t} = \pi_0 + \pi_1 UTSA_{i,t} + \pi_2 UTSA_Intensity_{i,t} + \pi_3 (UTSA_{i,t} \times DataIntensity_j) \\ + \Lambda^\top \mathbf{X}_{i,t} + \delta_t + \eta_j + u_{i,t}.$$

where $UTSA_{i,t}$ indicates post-adoption exposure in the firm's headquarters state and $DataIntensity_j$ measures industry-level data dependence. The instruments are strong, with first-stage F -statistics well above conventional thresholds.

In the second stage, predicted augmentation is used to explain integrated governance:

$$G_{I,i,t} = \alpha + \beta_{IV} \widehat{Augment}_{i,t} + \gamma^\top \mathbf{X}_{i,t} + \delta_t + \eta_j + \varepsilon_{i,t}.$$

The 2SLS estimates confirm a positive and economically large causal effect of augmentation on integrated governance. The IV coefficient is slightly smaller than, but statistically indistinguishable from, the baseline Logit estimates, suggesting that the main results are not driven by reverse causality or selection.

We further conduct a set of robustness checks. The results are stable to alternative constructions of augmentation and control-right measures, different governance thresholds, and alternative estimation approaches, including linear probability models and rank-normalized rights intensities. Across all specifications, augmentation remains positively associated with user-centric control rights and with the probability of integrated governance.

Overall, the empirical results closely mirror the theoretical mechanisms. As AI becomes more augmentation-oriented, firms reallocate control away from platform-centric access and exclusivity and toward rights that protect human–AI co-specialization. When contractual reallocation is insufficient, firms respond by integrating AI development within the firm boundary.

7 Conclusion

This paper develops an analytical, property-rights–theoretic framework for AI governance in which control is decomposed into a vector of separable digital rights—access, improvement, exclusivity, and forking—and shows how alternative allocations of these rights shape relationship-specific investment incentives under incomplete contracting. The model clarifies why AI adoption can generate chronic underinvestment in complementary intangible assets: when a platform retains residual control over a dynamic foundational model, the user firm anticipates ex-post holdup and (in centralized SaaS arrangements) an additional appropriation channel in which user-generated learning strengthens the platform’s outside option, disproportionately depressing incentives to invest in proprietary data. Intermediate governance structures mitigate distinct hazards through distinct mechanisms: exclusivity curbs appropriation by restricting the platform’s ability to redeploy user-derived value, while forking strengthens the user’s technological outside option but can weaken the platform’s incentive to invest in relationship-specific improvements. Full integration removes external contracting frictions at the cost of an internal developer-holdup problem when critical tacit knowledge cannot be fully codified into organizational capital.

To assess the empirical relevance of these theoretical mechanisms, we analyze a large corpus of firm-level AI disclosures from the annual 10-K filings of 3,482 publicly listed companies. Consistent with the model’s comparative statics, disclosures emphasizing augmentation-intensive AI applications are systematically associated with more user-centered governance structures, and discussions of exclusivity protections correlate strongly with greater investment in proprietary data assets. Taken together, these results suggest that observed heterogeneity in AI governance is not merely contractual noise but reflects economically meaningful attempts to reallocate control rights to protect relationship-specific investments and preserve complementarities in AI-enabled production.

This study shows that the dominant mode of AI provision in practice—platform-centered delivery through centralized SaaS arrangements (Regime C)—is not merely a neutral organizational choice, but one with systematic investment and welfare consequences. By

allocating residual control rights primarily to AI platforms, Regime C weakens firms' incentives to invest in complementary human capital and proprietary data, leading to chronic underinvestment in AI-enabled capabilities. As a result, equilibrium outcomes under the prevailing platform-based model generally fall short of the social optimum, even when AI technologies themselves continue to improve.

Our analysis highlights that this inefficiency is structural rather than accidental. When firms anticipate ex-post appropriation or lock-in by platforms, they rationally limit relationship-specific investments, constraining both private returns and aggregate welfare. Importantly, these distortions cannot be fully resolved through private contracting alone, as the relevant control rights are often non-contractible or weakly enforced.

The findings point to a clear role for public policy. By strengthening legal protection for AI users' control rights—such as enforceable rights to model improvement, data usage, or technological portability—or by clarifying property rights over certain forms of human capital and proprietary data investments, regulators can mitigate underinvestment incentives and restore complementarities in AI production. More broadly, the results suggest that effective AI governance should not focus solely on platform regulation or competition policy, but also on the institutional design of control rights that shape firms' incentives to invest. Rebalancing these rights has the potential to improve both firm-level outcomes and overall social welfare in AI-driven economies.

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Table 1: Marginal Private Investment Incentives with Asymmetric Bargaining Power (b)

Investment	Party	Regime	Marginal Private Return (Ignoring Costs)
AI Quality (A)	Platform (P)	C (Centralized)	$bV_A(A, L, D) + (1 - b)\phi V_A(A, \bar{L}, \bar{D})$
		U (User-control)	$bV_A(A, L, D) + (1 - b)\phi V_A(A, \bar{L}, \bar{D})$
		D (Distributed)	$b\epsilon V_A(A, L, D) + (1 - b)\phi V_A(A, \bar{L}, \bar{D})$
		I (Integrated)	$b(1 - \delta)V_A(A, L, D)$
Human Capital (L)	User (F)	C (Centralized)	$(1 - b)V_L(A, L, D) + bV_L(0, L, D)$
		U (User-control)	$(1 - b)V_L(A, L, D) + bV_L(0, L, D)$
		D (Distributed)	$(1 - b\epsilon)V_L(A, L, D)$
		I (Integrated)	$(\delta + (1 - b)(1 - \delta))V_L(A, L, D)$
Data Quality (D)	User (F)	C (Centralized)	$(1 - b)V_D(A, L, D) + bV_D(0, L, D)$
		U (User-control)	$(1 - b)V_D(A, L, D) + bV_D(0, L, D)$
		D (Distributed)	$(1 - b\epsilon)V_D(A, L, D)$
		I (Integrated)	$(\delta + (1 - b)(1 - \delta))V_D(A, L, D)$

Table 2: Main Empirical Results

Dependent variable	Augmentation coefficient	Interpretation
Right to Improve (r_{improve})	0.082***	Greater customization and adaptation rights
Right to Fork (r_{fork})	0.128***	Stronger technological autonomy
Right to Access (r_{access})	-0.098***	Reduced reliance on pure SaaS arrangements
Right to Exclusivity ($r_{\text{exclusive}}$)	-0.111***	Exit from external contracting space
Integrated Governance (G_I)	2.561***	Odds ratio ≈ 13 , indicating strong integration incentives

Notes: Coefficients report the effect of lagged AI augmentation. All regressions include firm-level controls, year and industry fixed effects. Standard errors are clustered at the firm level. *** $p < 0.01$.